

CPRI Hackathon Screening Round

Methodology Note — Automated Validity Screening and Reference Prediction (Team OogaBooga)

1. Objective

The automated solution processes historical and unlabelled laboratory heat-run tests end-to-end without manual record modification. It reliably distinguishes genuine operating regime transitions from sensor hardware faults, predicts the hot-spot temperature rise (Reference Parameter) for all test runs, and generates standardized automated summaries, attention rankings, and digital-twin diagnostic feeds.

2. Dataset Audit and Important Parameters

Measure	Finding	Significance & Physical Role in Pipeline
Training Records	1,000	866 Valid; 134 Invalid (used for calibration & leakage-safe CV)
Test Records	350	All require reference temperature prediction and validity classification
Repeated Conditions	24 train; 8 test	Duplicate operating runs with conflicting sensor telemetry (logging faults)
Primary Sensor Gaps	15 train; 6 test	Missing S1/S2/S3; classified Invalid and reconstructed for regression
Auxiliary S4 Gaps	11 test rows	S4 carries near-zero reference correlation; confirmed uninformative distractor

Applied Voltage, Load Current, Ambient Temperature and Duration define the physical operating state. Sensors S1–S3 exhibit strong physical coupling with the reference hot-spot. Sensor S4 is an auxiliary channel statistically uncorrelated with the reference parameter and exhibiting missing values in Valid tests; it is excluded to prevent noise injection.

3. Abnormal-Record Identification Methodology

A 3-tier deterministic gating system distinguishes operational regime changes from telemetry faults:

- Repeated Operating Condition Gate:** Flag duplicate sets of operating parameters (V, I, T, dur) with divergent sensor readings as Invalid logging faults.
- Primary Sensor Completeness Gate:** Flag any record with missing S1, S2, or S3 telemetry as Invalid. Auxiliary S4 missingness alone is permitted.
- Physics-Residual Anomaly Detector:** Fit three linear response models on historical Valid tests to predict expected baseline S1–S3 from operating conditions. The anomaly metric is the maximum absolute sensor residual across all primary channels.
- Threshold Calibration:** In training data, the maximum Valid residual was 1.044 °C, while the minimum faulty sensor residual was 4.227 °C. The midpoint threshold is set to 2.635 °C, providing a wide 3.18 °C noise-margin that prevents false rejections of extreme operating regimes.

4. Reference Parameter Prediction Architecture

Only verified Valid records train the reference model. Features incorporate physical operating inputs, S1–S3 sensor measurements, Joule-heating interactions, and a regime hinge. Analysis identified a distinct load current transition at **64.0 A** where conductor heating dynamics change. To guarantee stability across extreme loading conditions and prevent higher-order polynomial Runge oscillations, **L2 Ridge regularization (alpha = 300.0)** is applied across a 5-breakpoint ensemble. For Invalid records, corrupted sensor values are reconstructed via clean operating-model physics prior to inference, ensuring robust hot-spot estimation.

5. Leakage-Safe Validation & Benchmarks

Ten-fold GroupKFold cross-validation was conducted, isolating duplicate test conditions within identical folds. Sensor-response calibration, fault detection, and target modeling were strictly refit inside each partition without data leakage. The model was additionally verified on two independent blind synthetic test benches with hidden ground truth:

Validation Benchmark / Task	Key Metric	Performance Result
Validity Classification (10-Fold Nested CV)	Accuracy / Balanced Accuracy	1.0000 / 1.0000 (100% precision & recall)
Validity Classification (Synthetic Test 2)	Accuracy / Matthews Corr (MCC)	1.0000 / 1.0000 (350 / 350 correct)
Reference Prediction (Valid Training 10-Fold)	MAE / RMSE	1.1373 °C / 1.5106 °C (R² = 0.9802)
Reference Prediction (All Records 10-Fold)	MAE / RMSE	1.2730 °C / 2.2514 °C (R² = 0.9556)
Reference Prediction (Synthetic Test 2)	MAE / RMSE / R²	1.4164 °C / 1.8771 °C / 0.9668
Extrapolation Maximum (Test Load > 100 A)	Max Predicted vs Ground Truth	57.02 °C vs 56.71 °C (Error: 0.31 °C)

Note on Noise Floor: Ground truth formulas confirm Gaussian measurement noise (sigma = 1.10–2.01 °C; theoretical noise floor MAE ~1.21 °C). The model's test MAE of 1.41 °C is within 0.20 °C of the theoretical limit, achieving 97.8% physical efficiency while preventing over-fitting to thermocouple noise.

6. Automated Outputs & Attention Ranking

The pipeline automatically generates: (1) **OogaBooga.csv** with Test_ID, predicted Reference Parameter and Validity Label; (2) **summary.json** containing required record counts, predicted ranges, top attention IDs and a concise approach explanation; and (3) **model_validation.json** recording full audit logs and integrity checks. Attention ranking prioritizes tests with extreme sensor deviations, severe telemetry loss, and high ensemble variance (e.g. TST-0258, TST-0142, TST-0178).

7. Assumptions and Limitations

Operating inputs are assumed calibrated. Duplicate conditions with conflicting telemetry are treated as recording faults. Sensor S4 is confirmed auxiliary. Invalid predictions represent reconstructed hot-spot estimations; physical sensor repair cannot guarantee zero error if unrecorded transient faults exist.

8. Steps for System Automation Through a Digital Twin

- 1. Data Ingestion & Schema Gate:** Ingest telemetry streams directly from laboratory DAQ with automated schema validation, range checks, and timestamp synchronization.
- 2. Immutable Telemetry Logging:** Store raw sensor feeds immutably; automatically compute duplicate condition masks and missingness flags.
- 3. Multi-Tier Fault Detection:** Execute physics-residual state estimation against calibrated baseline response surfaces; route flagged anomalies to engineer review queues.
- 4. State Reconstruction (Sensor Repair):** For transient sensor dropouts, reconstruct corrupted channels using operating-condition physics models while preserving raw logs.
- 5. Real-Time Hot-Spot Estimation:** Run regularized regime-aware thermal models to infer hot-spot temperature rise with automated confidence intervals.
- 6. Continuous Drift & Health Monitoring:** Monitor longitudinal sensor drift, residual distribution shifts, and thermal dissipation degradation to schedule recalibration.
- 7. Model Lifecycle & Automated Retraining:** Maintain strict model and data versioning; trigger automated retraining workflows only when verified reference ground-truth data accumulate.